ADAS forecasting report

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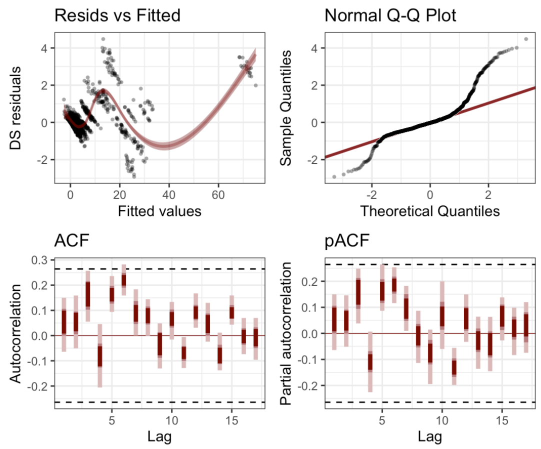
The ADAS contest aims to forecast four sets of outcomes. There are two pathogens, Yellow rust (*Puccinia striiformis*) and septoria leaf blotch (*Zymoseptoria tritici*). We then have two measurements for each pathogen, their crop incidence (CI) and their disease severity (DS). Crop incidence being the percentage of crops infected in a surveyed field, and disease severity being the percentage of an individual plant infected.

There are difficulties with this dataset. All measurements are percentages, which naturally leads to beta regression. CI and DS are on very different scales but are also directly correlated with each other. Most of the time series only show one or two peaks of infectivity, giving very little information to reconstruct the mechanism of the infection dataset.



I wanted to try a self-exciting process like a Hawkes process, unfortunately libraries for fitting them to time series are still rare. Instead, I focused on fitting a multivariate GAM using the *mvgam* library. In a multivariate process, I can model that the infection measures are driven by a shared underlying process. This lets me estimate their parameters precisely, leading to (hopefully) accurate predictions.

I wanted to see how well a statistical model would perform without mechanistic processes, so I only use the dataset given by the contest here. I ran into many difficulties using the different regions, so the end result is stripped back - a dynamic factor model estimated separately for CI and DS, with a shared 1-step autoregression as a latent trend. I did not include any covariates for this model. You can see in the residuals-vs-fitted graph above that there is a lot of structure in the residuals and in the QQ plot: likely the effects of hitting below 0 and above 100%. Careful prior selection may resurrect these results a little, but it is not obvious to me what scale these should be on.

Below are two hindcasts for two series. The model struggles with overfitting to some series, and outputs impossible negative results for other series. It models the overall dynamics well, but struggles to hit the largest extremes. My forecast for 2026 suggests that the disease burden in 2026 will be mild - unfortunate that this disagrees with some press releases. I suspect my model is underestimating the risk of future disease burden due to the long stretches of no disease between 2000 and 2020. I am not sure how better covariates could be collected to improve forecasting - mechanistic models would require both a within-population and between-region component which strikes me as very difficult. This was a tough challenge but one to which I was happy to try.